LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 2 Test: Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. A solution turns red litmus blue. Its pH is most likely to be: **[1]**

(a) 1 (b) 10
(c) 4 (d) 5

Q2. Which gas is given off when dilute hydrochloric acid is added to sodium carbonate? **[1]**

(a) Hydrogen (b) Chlorine
(c) Oxygen (d) Carbon dioxide

Q3. The chemical formula of bleaching powder is: **[1]**

(a) $CaCO_3$ (b) $Ca(OH)_2$
(c) $CaOCl_2$ (d) $CaSO_4$

Q4. Which of the following is used as an antacid? **[1]**

(a) Sodium chloride (b) Vinegar
(c) Milk of magnesia (d) Lemon juice

Q5. Plaster of Paris is obtained by heating: **[1]**

(a) limestone (b) baking soda
(c) washing soda (d) gypsum

Q6. **Assertion (A):** A concentrated acid should be diluted by adding the acid slowly to water, and not water to the acid. **Reason (R):** Mixing a concentrated acid with water is a highly exothermic process. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why does dry HCl gas not change the colour of dry blue litmus paper? **[2]**

OR

What is the difference between a strong acid and a weak acid? Give one example of each.

Q8. Write the chemical name and formula of baking soda. Write the equation for the reaction that takes place when it is heated. **[2]**

SECTION C

6 marks

Q9. A student tests three solutions P, Q and R with universal indicator and finds their pH to be 2, 7 and 12 respectively. **[3]**

(i) Which solution is neutral? **[1]**

(ii) Which solution has the highest concentration of H^+ ions? **[1]**

(iii) Which solution will turn phenolphthalein pink? **[1]**

Q10. What is the chlor-alkali process? Write the balanced equation and give one use of each product. **[3]**

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is a neutralisation reaction? Give one balanced equation. **[2]**

(ii) How is washing soda obtained from sodium carbonate? Write its formula and give one use. **[2]**

(iii) Why should curd and sour substances not be kept in brass or copper vessels? **[1]**

OR

(a) What happens when an acid reacts with a metal? Explain with a balanced equation, and describe how you would test the gas formed. [3]
(b) How do antacids give relief during indigestion? Name one antacid. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Tooth enamel, made of calcium hydroxyapatite (a form of calcium phosphate), is the hardest substance in the body. It does not dissolve in water, but it is corroded when the pH in the mouth falls below 5.5. Bacteria in the mouth produce acids by breaking down sugar and food particles that remain after eating. Toothpastes, which are generally basic, are used to clean the teeth. **[4]**

(i) Below what pH does tooth decay start? **[1]**

(ii) Why are toothpastes generally basic? **[1]**

(iii) What produces the acid in the mouth? Suggest two ways to prevent tooth decay. **[2]**

~ All the best! ~